1 An object of mass 1.5 kg is released from a stationary hot air balloon. Fig. 1.1 shows how the velocity of the object varies with time.

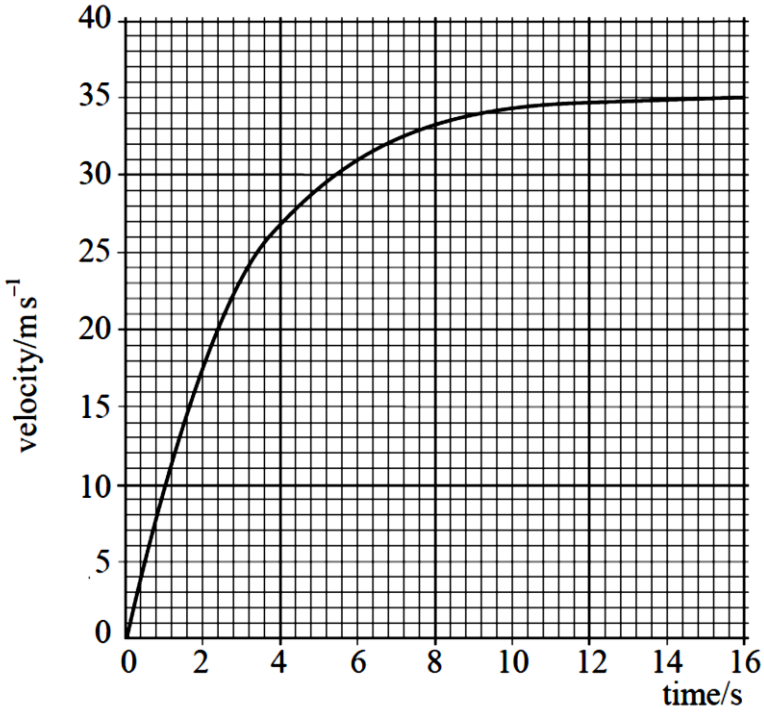

Fig. 1.1

(a) Describe how the acceleration of the falling object changes over the first 16 s.

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..... [2]

(b) Explain how the forces acting on the falling object cause the changes in acceleration described in **(a)**.

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..... [2]

(c) Use Fig. 1.1 to determine the acceleration of the object 5.0 s after it was released.

acceleration = …… m s^{-2} [2]

(d) Estimate the distance fallen in the first 16 s.

distance = …… m [2]

(e) Sketch the graph of distance against time using the axes provided below.

[2]